IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 30 April 2025, accepted 22 May 2025, date of publication 27 May 2025, date of current version 4 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3574210

RESEARCH ARTICLE

HistoDX: Revolutionizing Breast Cancer Diagnosis Through Advanced Imaging Techniques

WISHAL ARSHAD[1,2], TEHREEM MASOOD[2,3], H. M. SHAHZAD[2,3], HASSAN A. AHMED[4], SYED HAMZA AHMED[4], AND HAFIZ MUHAMMAD TAYYAB KHUSHI[2,3]

[1]Faculty of Software Engineering and Information Technology, University of Central Punjab, Lahore 54000, Pakistan
[2]Faculty of Computer Science and Information Technology, Superior University, Lahore 54000, Pakistan
[3]Department of Software Engineering, Superior University, Lahore 54000, Pakistan
[4]Department of Information Systems, Monte Ahuja College of Business, Cleveland State University, Cleveland, OH 44114, USA

Corresponding author: H. M. Shahzad (shahzad.dar@superior.edu.pk)

ABSTRACT Breast cancer is the second leading cause of mortality among women worldwide, highlighting the need for efficient histopathology-based screening methods for early diagnosis. This study introduces HistoDX, a deep learning framework to classify Invasive Ductal Carcinoma (IDC) using 277,524 histopathology image patches (50 × 50 pixels) from Paul Mooney's IDC dataset on Kaggle, comprising No Cancer and IDC(+) classes. HistoDX employs a preprocessing pipeline with normalization, data augmentation, and class balancing via oversampling and weighted loss to address the class imbalance. A customized convolutional neural network, built on EfficientNetV2-B3 with additional layers, achieves 97% accuracy and a 0.91 ROC-AUC score on the test set. Validation on BreakHis (97% accuracy) and BACH (90% accuracy) datasets confirm generalizability, though detecting under sampled IDC(+) cases remains challenging. Low training and test losses underscore reliability. HistoDX empowers pathologists by enhancing diagnostic efficiency and minimizing subjectivity through effective class imbalance mitigation. Its robust performance across diverse datasets like BreakHis and BACH suggests readiness for clinical integration. Future research into advanced augmentation techniques, ensemble models, and whole-slide image analysis could further optimize accuracy, sensitivity, and scalability, paving the way for broader adoption in precision oncology.

INDEX TERMS Breast cancer, invasive ductal carcinoma, EfficientNetV2-B3.

I. INTRODUCTION

Cancer is the result of an aberration in the typical regulation of cellular proliferation, which is instigated by mutations in the cellular genetic material. In a physiological state, the processes of cellular growth, division and apoptosis occur in a meticulously regulated fashion. However, the accumulation of genetic damage or mutations disrupts this regulatory framework. Such genetic alterations may lead to Oncogene activation to encourage uncontrolled cell division or the inactivation of tumour suppressor genes, which regulate cell growth and facilitate DNA repair. It additionally contributes to the uncontrollable growth of abnormal cells. Tumours are created when aberrant cells come together to form a solid mass. There are two types of tumours: benign, which are not cancerous and malignant, which tend to spread to other tissues. Metastatic potential is the tendency of malignant tumours to spread through the lymph or blood vessels to different parts of the body and it is one of the characteristics that makes cancer so lethal. The risk of developing cancer is increased by several factors, including genetic predispositions, personal lifestyle decisions and environmental exposures to cancer-causing substances [1].

Breast cancer's global impact has become an essential issue for the community. The most widespread form of cancer is breast cancer in adults, with over 2.5 million new cases diagnosed yearly. Breast cancer is also the first or second foremost cause of mortalities due to cancer for women in 95% of countries. This problem is especially noticeable in low and middle-income areas, where late diagnoses and constrained

The associate editor coordinating the review of this manuscript and approving it for publication was Andrea F. Abate.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

access to healthcare resources have a negative impact on survival rates. Initiatives like the Global Breast Cancer Initiative Framework from the World Health Organization are being implemented in response to this terrible circumstance. The primary types of breast cancer are invasive and non-invasive; non-invasive cancers, also known as carcinoma in situ, are only present in the ducts or lobules and yet to spread to surrounding tissues, while invasive breast cancer extends beyond the ducts or lobules into surrounding tissues and is more likely to metastasize. Out of all the invasive breast cancers, IDC is the most common, making up approximately 70 to 80 per cent of all cases; IDC starts in the milk ducts, as illustrated in Fig. 1, but can spread to other parts of the breast tissue to distant organs. It is typically classified into various subtypes based on the appearance of the cells, hormone receptor status and HER2 status, which are critical for determining prognosis and treatment strategies. IDC presents a challenging clinical problem due to its ability to metastasize, often detected in later stages when the cancer has spread beyond the breast. The molecular and genetic alterations that drive IDC remain an area of active research, focusing on understanding how mutations in genes like TP53, BRCA1/2 and others contribute to tumour progression and resistance to therapy [2].

Immediate and credible breast cancer diagnosis is essential for improving patient outcomes and survival rates [3]. Early detection using multiple imaging techniques makes timely intervention possible, enhancing the possibility of a successful course of treatment. Mammography, ultrasound, computed tomography (CT) and magnetic resonance imaging (MRI) are examples of these non-invasive imaging methods that are essential for identifying irregularities in breast tissue [4], [5]. Mammography and ultrasound are the most commonly used approaches to initial examinations. Even with the continued development of imaging technologies, histopathological analysis of biopsy samples remains the gold standard for the accurate identification and classification of breast cancer types. This guarantees the highest level of correctness when deciding the tumour type and directing options for therapy. Combining these techniques for diagnosis can help medical professionals give patients better care and increase survival rates [6].

Histopathology, which employs microscopic tissue analysis to differentiate benign from malignant growths, is necessary for the diagnosis of breast cancer (BC). It helps alongside classification, staging and treatment planning by offering essential details about tumour morphology. It remains the gold standard for precise diagnosis regardless of improvements in imaging and AI technologies. Histopathology is going through a revolution. It's because deep learning automates image analysis while improving precision and decreasing errors. These advancements strengthen patient outcomes and the speed of diagnosis [7].

In clinical settings, pathologist's manual and qualitative evaluation remains the primary approach for histopathological analysis. However, this method faces significant challenges. First, there is a worldwide shortage of pathologists, particularly in low-resource settings and smaller hospitals, where the pathologist-to-patient ratio can be as low as 1:100,000 in some regions [8]. This scarcity delays timely diagnoses, exacerbating patient outcomes. Moreover, the accuracy of histological diagnosis depends heavily on the pathologist's skill and experience, introducing subjectivity that can lead to diagnostic discrepancies. Studies suggest inter-observer variability rates as high as 20–30% for complex cases like IDC. The heavy workload, with pathologists often reviewing hundreds of slides daily, further compounds these issues, heightening the risk of fatigue-induced errors. Overcoming these challenges, particularly in classification tasks, necessitates automated, precise histopathological image analysis tools powered by deep learning to minimize human error, enhance objectivity and boost diagnostic efficiency [9].

Machine learning and image processing algorithms offer helpful information for breast cancer diagnosis and detection. However, the frequent occurrence of false positives and false negatives could affect the reliability of these techniques. Convolutional neural networks (CNNs), a cornerstone of deep learning technologies, demonstrate promise in the early detection of breast cancer by reducing the time required for mammography interpretation. Beyond speed, CNNs offer a transformative histopathological solution by automating slide analysis, alleviating the burden on overworked pathologists and providing objective, reproducible classifications to counter subjectivity. These capabilities position deep learning as a vital tool for assisting medical professionals in resource-constrained environments, potentially enhancing diagnostic consistency and patient care. Even though there are numerous deep-learning algorithms, not all of them have been thoroughly tested for the identification of breast cancer. By extracting meaningful information from images, these algorithms may complete fully automated tasks like mass segmentation, detection and classification without human intervention. One problem, though, is that deep CNNs cannot be trained efficiently when there are not enough images.

Transfer learning (TL), which uses pre-trained models to improve diagnostic performance, is crucial in these situations. This is especially visible in industries like medical image analysis, where complex procedures are required to analyze images like breast scans. Recently, researchers have concentrated on strengthening transfer learning networks by employing pre-trained weights to improve classification tasks and increase interpretation accuracy. The present study employs an advanced algorithm to detect breast cancer using histopathology images. Many of the images in the collection are classified as No Cancer or IDC(+). The training set has 265,142 images, whereas the validation set has 128,382. To ensure quality, each image was shrunk to 50 by 50 pixels. This study concentrates on an EfficientNetV2-B3 model for detecting breast cancer, which is noteworthy for its

high accuracy but low computational cost. Experiments have shown that computation time and memory usage are significantly reduced through the optimized design of this model's architecture. Still, it can also identify the IDC commonly found in large datasets such as breast cancer whole-slide images (WSIs). The most common subtype of breast cancer is IDC, which calls for prolonged research to improve the precision of diagnosis. We utilized the Breast Histopathology Images (IDC Dataset), which was compiled by Paul Mooney and made accessible on Kaggle, for this investigation. Deep learning-based diagnostic models can be created and evaluated thanks to this dataset's extensive collection of high-quality, annotated image patches. We utilized the Breast Histopathology Images (IDC Dataset), compiled by Paul Mooney and made accessible on Kaggle, for this investigation. Deep learning-based diagnostic models can be created and evaluated thanks to this dataset's extensive collection of high-quality, annotated image patches. While this dataset provides a robust foundation for model development, its single-source nature highlights the need for future validation across diverse, multi-institutional datasets to ensure broader clinical applicability. Retraining using transfer learning can effectively solve the problem of disuniformity in medical image recognition caused by a limited number of annotated data, so its efficiency and overall accuracy have improved beyond recognition. Through employing this technique, the study demonstrates that EfficientNetV2-B3 can help pathologists and contribute towards improved screening [10].

To ensure the model's robustness beyond a single-source dataset, this study further evaluates HistoDX on BreakHis dataset and BACH dataset assessing its performance across varied imaging conditions and preprocessing methods. This multi-dataset validation complements the primary analysis on the IDC dataset, offering insights into the model's generalizability and potential for clinical deployment across diverse scenarios. This research aims to create an effective automated deep-learning system to help radiologists correctly classify histopathology images. The research's objectives and methodology are as follows:

- To ensure uniform input dimensions for the model, the images were resized to 50 × 50 pixels using bilinear interpolation (`cv2.INTER_LINEAR`). During the preprocessing stage, the pixel values were normalized to the range [0, 1] to standardize the input data, accelerate model convergence and boost computational efficiency.
- Class imbalance was dealt with by oversampling the under sampled class IDC(+) by employing a weighted loss function that discourages misclassifications with greater severity by providing the under sampled class higher weights.
- To ensure an adequate model evaluation, the dataset has been separated into subsets for testing, validation and training.
- With pre-trained weights on ImageNet enabling extensive feature extraction capabilities, the EfficientNetV2-B3 model was chosen for the breast cancer classification work because it strikes the ideal balance between accuracy and computational economy.
- The model's performance on the training and validation datasets was evaluated using accuracy, precision, recall and F1 score, providing an in-depth assessment of its classification abilities.

The main aim was to create an automated tool to help radiologists categorize breast cancer tumours using the EfficientNetV2-B3 model to improve classification accuracy.

The structure of the paper is as follows: Section II provides an informative summary of important studies on the most innovative techniques and breakthrough developments in medical image analysis. Section III illustrates the suggested approach, which incorporates data enhancement and microscopic image processing. It describes the suggested deep learning-based architecture and transfer learning paradigm. Section IV shows results and Evaluation parameters. Section V summarizes the research's findings in the proposed framework using microscopic images and provides a short overview of the results, emphasizing recent developments in deep learning and its applications in microscopic image interpretation. Lastly, Section VI concludes the findings of the study.

FIGURE 1. Invasive ductal carcinoma.

II. LITERATURE REVIEW

This section discusses prior studies on the utilization of CNNs as feature extractors for the diagnosis of breast cancer [11]. To forecast the image's label, a classifier receives the features that CNNs have extracted in several experiments [12]. Convolutional, pooling and fully linked layers are implemented to extract high-level properties from an image. This is the fundamental idea of CNN. Together,

these layers locate and extract meaningful information from the image, enabling more accurate detection and classification [13]. This section highlights the use of standard CNN designs and techniques to enhance their functionality and current developments in breast cancer diagnosis. The focus is on showcasing previous research pertinent to the methodology suggested in this investigation. Studies have employed architectures like ResNet, DenseNet and VGG, but EfficientNet variants, particularly V2-B3, stand out for balancing high accuracy with computational efficiency, a critical advantage for histopathology analysis.

Leslie Solorzano et al. have developed an ensemble deep-learning model to detect invasive carcinoma (IC) in breast cancer whole slide images. The study used 587 WSIs, achieving improved accuracy and dice score through ensemble learning, also allowing for examining base-model agreement. Inception V3 was selected based on its strong performance in prostate cancer and results were aggregated using majority voting. The model achieved an accuracy of 92.7%. This methodology, focused on binary classification, also highlighted the importance of data sampling and model selection. Future work aims to explore explainability and multi-class classification [14].

Beyond traditional CNNs, recent studies have advanced feature extraction techniques, offering higher quality and robustness compared to earlier literature. For instance, fuzzy logic-integrated CNNs enhance resilience in noisy medical images [15], as shown in a 2024 IEEE Transactions on Fuzzy Systems study, which proposes a hybrid model improving performance on complex histopathology datasets like those in breast cancer analysis, potentially surpassing standard EfficientNetV2-B3 in challenging conditions. Similarly, advanced segmentation models like Mask R-CNN outperform traditional architectures in medical imaging [16], with a 2022 Frontiers in Public Health study demonstrating its effectiveness in precisely segmenting the left ventricle in echocardiography images, achieving a Dice Similarity Coefficient (DSC) of up to 0.9221 and accuracy of 0.9876, surpassing conventional CNNs like SegNet and FCN. Additionally, a 2023 Frontiers in Bioengineering and Biotechnology study applies CNNs to bioinformatics [17], likely enhancing precision in tasks such as breast cancer biomarker identification, complementing EfficientNetV2-B3's efficiency with domain-specific insights. These innovations contextualize our use of EfficientNetV2-B3, which achieves 96.3% accuracy [18] but lacks the specialized robustness of fuzzy integration, the precision of advanced segmentation, or bioengineering adaptations, suggesting avenues for future enhancement.

Using the BreakHis dataset, Jacinta Potsangbam delivered a deep learning-based method for performing binary classification of breast cancer histopathology images. The framework operates on a DenseNet-121 pre-trained model for feature extraction and data augmentation for preprocessing. The fully connected layers (FCL) further refine the model. The 100X magnification factor resulted in an optimal accuracy of 96.09%. Compared to previous investigations, experimental results showed increased accuracy for all magnification factors. This method describes how transfer learning might enhance the use of histopathology images to determine the presence of breast cancer [19].

Kumbhare et al. proposed a novel breast cancer detection model leveraging Federated Learning (FL) to enhance diagnostics without centralizing data, utilizing mammography images from the Digital Database for Screening Mammography (DDSM) dataset. The model employs DenseNet for feature extraction and Enhanced Recurrent Neural Networks (E-RNN) for classification, optimized with a Hybrid Dragon-Rider Optimization (HDRO) algorithm combining Dragonfly and Red Deer techniques, achieving 95% accuracy and 91% MCC [20]. In a broader context, Yi et al. reviewed 150 studies on Generative Adversarial Networks (GANs) in medical imaging, highlighting their prominence in histopathology tasks like breast cancer detection. GANs excel in data augmentation to address data scarcity and class imbalance, generating synthetic images to improve model generalization unlike Kumbhare's DenseNet reliance, which lacks such augmentation. Compared to EfficientNetV2-B3's efficiency in [18], GAN-enhanced models offer robustness but demand higher computational resources. Yi et al. also note GANs' potential for interpretability via learned priors from discriminators, though specific methods like Grad-CAM remain underutilized, mirroring a gap in Kumbhare's work [21].

Singh et al. proposed a hybrid model leveraging transfer learning for breast cancer detection using histopathological images, aiming for early and cost-effective diagnosis. The study combined deep neural networks with machine learning classifiers, where DenseNet + Logistic Regression achieved an F-score of 0.81, outperforming VGG + Logistic Regression (0.73), VGG + Random Forest (0.74) and DenseNet + Random Forest (0.79), validated across histopathological datasets [22]. In contrast, Talaat et al. introduced BCaXAI, a 3D Inception-ResNet V2 model with Grad-CAM for mammogram-based breast cancer classification, achieving 98.53% accuracy and an F1-score of 0.9843 on the CBIS-DDSM dataset. Unlike Singh's DenseNet approach, BCaXAI emphasizes interpretability, using Grad-CAM heatmaps to highlight tumour regions, enhancing radiologist trust. That is an advance over EfficientNetV2-B3's 96.3% accuracy [18] without such visualization. While Singh's model lacks explicit class imbalance handling, BCaXAI ensures balanced training data curation, surpassing our study's oversampling in robustness, though at higher computational cost [23].

Singh et al. proposed a hybrid model leveraging transfer learning for breast cancer detection using histopathological images to enable early and cost-effective diagnosis. The study evaluated various combinations of deep neural networks and machine learning classifiers like F-score of 0.81, the DenseNet + Logistic Regression model outscored VGG

+ Logistic Regression (0.73), VGG + Random Forest (0.74) and DenseNet + Random Forest (0.79). Extensive experiments on histopathological image datasets validated the model's efficacy in detecting breast cancer with superior performance compared to baseline methods [22].

Hasan et al. proposed a unified histopathology image segmentation and classification architecture to enhance breast cancer diagnosis. Combining SegNet and U-Net for segmentation with ResNet, VGG and DenseNet for classification and using Stardust as the backbone, the model achieved a classification F1-score of 0.833 on a private dataset, with segmentation F1-scores of 0.902 (ER-IHC-stained private dataset) and 0.903 (HE-stained MoNuSeg dataset). Real-time testing on PR-IHC-stained slides showed robustness across 20x and 40x magnifications, supporting rapid pathology workflows [24]. Compared to EfficientNetV2-B3's 96.3% accuracy in IDC detection [18], Hasan's approach prioritizes segmentation alongside classification, though its lower F1-score suggests trade-offs in classification precision. Unlike Zarif's EfficientNetV2-B3 study, which lacks interpretability tools [18], Hasan's work omits Grad-CAM or similar methods, a gap also evident in Singh's DenseNet study [22]. For class imbalance, Hasan relies on dataset diversity rather than explicit techniques like our oversampling or GANs noted by Yi et al. [21], potentially limiting generalizability.

Muhammad Mumtaz Ali introduced ECAM, an Enhanced Channel-Wise Attention Mechanism, for improved diagnosis of IDC using histopathological images. The study focuses on enhancing computational efficiency with a Separable CNN architecture, hierarchical feature aggregation and attention mechanisms. Evaluated on DataBioX IDC and BreakHis datasets, ECAM surpassed state-of-the-art models like DenseNet121, VGG16 and AlexNet. On IDC data, ECAM achieved an accuracy of 96.70% and F1 score of 96.65%, while on BreakHis, it attained 96.33% accuracy and 96.37% F1 score. This model offers significant potential for automated IDC grading [25].

Karan Gupta's research focuses on enhancing breast cancer diagnosis using automated histopathology image analysis. The study retrieves activation traits using CNNs and evaluates how they perform when compared with additional conventional classifiers like SVM and Logistic Regression. A two-phase model was proposed to classify images by magnification levels (40x, 100x, 200x, 400x) and detect benign versus malignant tumours. The dataset was split into 80% for training and 20% for testing. Experimental results revealed that ResNet50 with LR achieved the highest accuracy of 95.2%, outperforming CNN + SVM. This model enhances feature extraction and improves classification accuracy, aiding automated diagnosis systems [26].

Li et al.'s study introduces a novel convolutional neural network (CNN) architecture for classifying benign and malignant breast cancer (BC) in histological images. The architecture integrates DenseNet as the base with interleaved Squeeze-and-Excitation (SENet) modules to enhance feature delivery and utilization. Extensive experiments conducted on the publicly available BreakHis dataset demonstrate that the proposed framework significantly outperforms state-of-the-art CNN methods, achieving an accuracy of 97.1% in BC classification, underscoring its potential for improving diagnostic precision [27]

Wang et al.'s study addresses the challenges of diagnosing and classifying Breast Cancer (BC), particularly IDC, using a hybrid deep learning model (CNN-GRU). The model combines CNN layers with Gated Recurrent Units (GRU) to analyze Whole Slide Images from the PCam Kaggle dataset, automating IDC detection. It achieved promising results, including an accuracy of 86.21%, precision of 85.50% and an F1-score of 88%, outperforming models like CNN-LSTM and CNN-BiLSTM. With an AUC of 0.89, the model demonstrates robustness in overcoming pathologist errors and misclassification. These findings emphasize its potential for improving early and accurate BC detection. The proposed hybrid model proves to be more effective than several recent ML/DL approaches [28].

Barsha et al. presented ensemble deep-learning models for the detection and grading of IDC and breast cancer. For IDC detection, both DenseNet-121 and DenseNet-169 network architectures were utilized with test time augmented (TTA) data to obtain 92.70% balanced accuracy and 95.70% F1-score. The pathological grading of IDC was tackled with an ensemble of DenseNet-121, DenseNet-201, ResNet-101v2 and ResNet-50, achieving patch-level classification accuracies of 75.07% (10×) and 79.14% (20×), respectively. These findings suggest that a model can help pathologists identify and grade IDC to help in more informed treatment plans [29].

Choudhary et al.'s study addresses the challenges of deploying deep learning models for breast cancer diagnosis on mobile and low-cost devices. A novel approach combining transfer learning with structured filter pruning is proposed to classify histopathological images while reducing computational demands. Filters deemed less critical are pruned, followed by retraining the models. Using pre-trained CNNs, the pruned VGG19 model achieved 91.25% accuracy with a 63.46% reduction in FLOPs. The ResNet34 pruned model improved accuracy to 91.80% while cutting FLOPs by 40.63% and ResNet50 pruned model achieved 92.07% accuracy with a 30.97% reduction in FLOPs. This method demonstrates efficiency and accuracy, making it suitable for resource-constrained devices [30].

Zeng et al.'s research explores the feasibility of using AutoML technology to detect IDC in whole slide images. The study employs Google Cloud AutoML Vision by passing handcrafted neural networks and using an augmented dataset of 378,215 labelled histopathology images to balance IDC(+) and No Cancer. The proposed model achieves 91.6% accuracy (AuPRC) during evaluation and 84.6% balanced accuracy on a held-out test set, outperforming previous studies. Generalization tests on new samples collected from hospitals confirm similar performance. The findings highlight the potential of AutoML, especially when paired

TABLE 1. Compilation of research studies on breast cancer detection.

Author Name (Reference)	Dataset	Model(s) Used	Accuracy (%)	Limitations
Leslie Solorzano et al. [14]	587 WSIs	Inception V3	92.7%	Lacks interpretability
Jacinta Potsangbam [19]	BreakHis	DenseNet-121	96.09%	No imbalance handling
Savita Kumbhare [20]	DDSM	DenseNet, E-RNN, HDRO	95.0%	No Grad-CAM
Sukhendra Singh [22]	IDC Patches	DenseNet	86.97%	Lacks interpretability
Md Jahid Hasan [24]	ER-IHC, MoNuSeg	U-Net, SegNet, ResNet, VGG, DenseNet	F1: 0.833 (class.), 0.902 (seg.)	No explicit imbalance fix
Muhammad Mumtaz Ali [25]	DataBioX IDC, BreakHis	ECAM	96.7% (IDC), 96.33% (BreakHis)	High efficiency
Karan Gupta [26]	BreakHis	ResNet50	95.2%	No XAI
Xia Li [27]	BreakHis	DenseNet + SENet	97.1%	Computationally heavy
Xiaomei Wang [28]	PCam Kaggle	CNN-GRU	86.21%	Lower accuracy
Nusrat Ameen Barsha [29]	Not Specified	DenseNet-201	92.7%	No interpretability
Tejalal Choudhary [30]	IDC	ResNet50	92.07%	No imbalance focus
Yan Zeng [31]	IDC	AutoML Vision	91.6%	Lacks transparency
Yusuf Celik [32]	IDC	DenseNet-161, ResNet-50	94.11%	Resource-intensive
Mohammad Abdolahi [33]	IDC	CNN, VGG-16	85%	Basic approach
Selina Sharmin [34]	IDC	ResNet50V2, LGB	95%	Some interpretability
Jianfei Zhang [35]	WSIs	MSRCNN, SVM	87.45%	Moderate accuracy
Nassib Abdallah [36]	BreakHis, IDCDB	Feature Pipeline	95.2% (BreakHis), 94.7% (IDCDB)	No DL interpretability
Aditi Kajala [37]	Histopathology	EfficientNet	87.5%	Overfitting risk
Melwin D Souza [38]	IDC	VGG11, EfficientNet	91.53%	Balanced efficiency
Sruthi Krishna [39]	BreakHis	EfficientNet	96.36%	No XAI
Sania Eskandari [40]	IDC	Vision Transformer (ViT)	93%	Resource-heavy
Abhinav Kumar [41]	IDC WSIs	SLIDE-Net	89%	Imbalance addressed
M. Iffat Hossain [42]	IDC	ResNet-50, Inception V2, EfficientNet	94.05%	No Grad-CAM
Masoud Tafavvoghi [43]	H&E WSIs	XGBoost	96.5%	No DL visuals
Sameh Zarif [18]	IDC	CNN, EfficientNetV2-B3	96.3%	Lacks interpretability

with cloud computing, for scalable IDC detection in digital pathology [31].

ResNet-50 and DenseNet-161 were utilized in Yusuf Celik's study to automatically identify IDC in histopathology images using AI and transfer learning. DenseNet-161 got 92.38% and 91.57%, respectively, whereas ResNet-50 got an F-score of 94.11% and balanced accuracy of 90.96%. The models were also validated on the BreakHis dataset, demonstrating high performance in classifying histopathology images across magnifications into benign and malignant classes. These promising results establish the system's potential for broader application with more diverse datasets [32].

Abdolahi et al.'s research focuses on leveraging deep learning for the early detection and classification of IDC in breast cancer. Using 277,524 image patches, the study compares two approaches: a baseline convolutional neural network (CNN) trained from scratch and transfer learning with the pre-trained VGG-16 model. The baseline model achieved an F-measure of 83% and an accuracy of 85%, outperforming prior studies on the dataset. Transfer learning with feature extraction showed better results than handcrafted features, achieving 81% for both F-measure and accuracy while also being more time-efficient than training from scratch. These findings emphasize the efficacy of deep learning, particularly transfer learning, for histopathology image analysis [33].

Sharmin et al.'s study shows that the model combines the feature extraction capabilities of deep learning with its interpretability and generative characteristics. To give a clear indication of observed results, rigorous experiments were performed using the website's available IDC dataset. The study excelled in state-of-the-art methods that acquire 95% accuracy with precision of 94.86%, recall of 94.32% and F1-score of 94.57%. The utilization of the Light Boosting Classifier (LGB) in conjunction with ResNet50V2 has shown potential for enhancing patient care and breast cancer detection [34].

For IDC identification, Zhang et al.'s study blends SVM with a multi-scale residual CNN (MSRCNN). Features have been identified and categorized according to preprocessed WSI patches. With 5-fold cross-validation, the approach got 79.89% F1-score, 87.45% accuracy and 85.7% balanced accuracy. This approach holds significant potential for practical applications and further research in automated breast cancer diagnosis [35].

In order to eliminate intra and inter centre variability challenges, Abdallah et al.'s paper presents a feature engineering-based pipeline for categorizing benign tumours and IDC in histopathological images. In order to reduce batch effects brought on by variations in image acquisition and a range of patient variables, the pipeline incorporates feature extraction and harmonization algorithms. The pipeline outperformed current feature engineering-based models and corresponding deep learning models, achieving high accuracy rates of 95.2% and 94.7%, respectively, when tested on the BreaKHis and IDCDB datasets. The results of this study provide proof of how vital harmonization is to enhance the performance and resilience of models [36].

Kajala et al.'s study explores the use of EfficientNet architectures (B0-B7) for breast cancer classification using histopathology images. EfficientNet-B2 achieved the highest performance with a validation accuracy of 0.875, an AUC score of 0.95 and a validation loss of 0.2945. Models B6 and B7 faced overfitting, which can be mitigated with additional techniques. The study highlights the potential of computer-aided diagnosis to assist pathologists and improve diagnosis efficiency [37].

Souza et al.'s research focuses on developing a novel transfer learning approach for early-stage breast cancer detection using VGG11 and EfficientNet architectures. The proposed model, named 'EARLYNET,' automates breast cancer prediction by distinguishing benign from malignant tumours, achieving an accuracy of 91.53%. The study highlights how combining these architectures can improve the accuracy, robustness and computational efficiency of early-stage breast cancer detection, ultimately aiding in earlier diagnosis and better patient outcomes [38].

The proposed system in Krishna et al.'s study achieved high performance using EfficientNet for breast cancer detection. Specifically, the model achieved an accuracy of 96.31% for 200x magnification images and 96.36% for 400x magnification images. This demonstrates EfficientNet's effectiveness in accurately classifying histopathology images, further supported by its efficiency in terms of test time, making it a strong candidate for high throughput breast cancer screening [39].

Eskandari et al.'s study focuses on comparing the performance of eight deep-learning models for classifying histopathological images and detecting breast cancer. These models include ResNet-50, DenseNet-121, ResNeXt-50, Vision Transformer (ViT), GoogLeNet (Inception v3), EfficientNet, MobileNet and SqueezeNet. The study found that the ViT model, with its attention-based mechanisms, achieved an impressive validation accuracy of 93%, outperforming traditional convolutional networks. This research underscores the potential of advanced machine learning techniques for enhancing the precision and efficiency of breast cancer diagnosis in clinical settings [40].

Kumar et al.'s research introduces SLIDE-Net, a framework for detecting IDC in Whole Slide Images. The model integrates SequentialLSTM with EfficientNet to effectively capture contextual information across tissue image patches, addressing the challenges of data imbalance in IDC datasets using an Adaptive Fuzzy C-Mean approach. SLIDE-Net demonstrated excellent performance, achieving 86% accuracy, balanced accuracy, sensitivity, specificity and 89% accuracy on the balanced PatchCAM dataset. These results highlight the model's capability to utilize contextual patch information for superior IDC detection, offering a robust and reliable solution [41].

Hossain et al.'s research focuses on detecting IDC using an ensemble of deep learning models ResNet-50, Inception V2 and EfficientNet leveraging transfer learning. The study uses a dataset of 162 whole-mount slide images, from which 277,524 50 × 50 image patches were extracted (including both IDC(+) and No Cancer patches). By employing 10-fold cross-validation, the ensemble method achieved an accuracy of 94.05%, surpassing the individual model performances. The ensemble approach also demonstrated improved performance across other metrics, offering a promising solution for IDC detection in breast cancer diagnosis [42].

Tafavvoghi et al.'s study eliminates subjective IHC and expensive gene profiling through the use of H&E-stained WSIs for breast cancer molecular subtyping. Deep learning's guarantee for molecular subtyping was demonstrated by a two-step pipeline that got 96.5% accuracy, an F1 score of 0.95 for tumour classification and a macro F1 score of 0.73 for subtyping [43].

Zarif's hybrid CNN+EfficientNetV2-B3 model achieved 96.3% accuracy for IDC detection [18]. Compared to ResNet-50 (94.11% [32]) or DenseNet-161 (92.38% [32]), EfficientNetV2-B3's optimized scaling reduces computational demands, making it suitable for large-scale WSI analysis while maintaining high performance.

The limitations of these methods are discussed in the comparison Table. 1 and literature review. It highlights the drawbacks of current studies on deep learning-based breast cancer evaluation, emphasizing a number of major problems. A major issue is the small size and absence of diversity of datasets, which might impact how well models generalize to other datasets or real-life circumstances. Class imbalances and biases related to magnification are prevalent in datasets that include BreakHis, IDC and DDSM, which may restrict the ability of the model to generalize. Furthermore, many deep learning models have a problem of overfitting, especially if they are trained on restricted datasets or high-magnification images. The model's utility in clinical settings is limited in this overfitting, resulting in poor generalization to data that is not known.

The computational expense of many models, particularly deep architectures like ResNet-50 and DenseNet-161, is another critical disadvantage. These models are less practical for deployment in situations with limited resources, like mobile devices or hospitals with low computer power, because they demand significant resources for both training and inference. Furthermore, as many deep learning models do not offer transparent decision-making processes, model interpretability continues to be a problem. Medical professionals require models that not only perform successfully but also provide a reason for their choices, something that is frequently absent from previous studies.

Another problem with these models is their usefulness; although some of these perform well in controlled research settings, others have difficulty generalizing to clinical settings where patient diversity, imaging quality and preparation differ. Furthermore, there are still problems with scalability and integration. Many models are not suited for seamless integration in clinical processes. Rarely are aspects like integration with medical systems and real-time processing examined. The accuracy and consistency of models across

many institutions are made more difficult by variances in data quality, staining methods and magnification. Even while some research has attempted to address these issues, they continue to be significant barriers to the development of strong and reliable models. Lastly, two critical issues are possible biases in the data and the explainability of deep learning models. In medical applications, where transparency and trust are essential, many models fail to offer clear reasons for their predictions. Furthermore, biased models that might not function as effectively for all patient populations can result from data imbalances, such as the underrepresentation of particular demographic groups. The proposed approach aims to address these weaknesses by emphasizing the need for better generalization, more diverse and high-quality datasets, and enhanced interpretability.

III. MATERIALS AND METHODOLOGY

This study develops HistoDX, leveraging EfficientNetV2-B3 to classify breast histopathology images. It achieves 97% accuracy and 0.91 ROC-AUC. This section details the dataset, preprocessing and customized model for breast cancer detection.

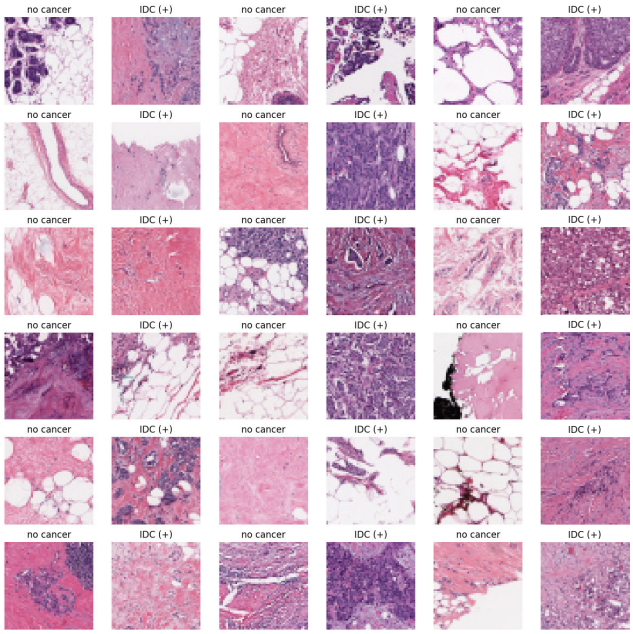


FIGURE 2. IDC(+) and no cancer histopathology images.

A. DATASET INFORMATION

We used Paul Mooney's curated Breast Histopathology Images (IDC dataset) from Kaggle. The dataset comprises 277,524 patches (50 × 50 pixels, 40× magnification) from 279 patients, derived from Invasive Ductal Carcinoma (IDC) slides [44], as shown in Table 2. It includes 78,786 IDC(+) and 198,738 No Cancer patches, stored in PNG format with naming convention `u_xX_yY_classC.png` (u z patient ID, X/Y = coordinates, C = class: 0 = No Cancer, 1 = IDC(+); Fig. 2). Spanning 279 patients, the dataset implicitly captures variations in age, pathology and tissue characteristics, though explicit demographic diversity (e.g., race) is not detailed. Significant class imbalance (71.6% No Cancer vs. 28.4% IDC(+)) was addressed using class weights during training to reduce bias toward the majority class. For robustness evaluation, the model was additionally tested on the BreakHis dataset and BACH dataset. These datasets, differing in source, preprocessing and characteristics, were used to assess generalizability without retraining, providing a broader validation scope beyond the primary IDC dataset. Detailed performance across these datasets is reported in Section IV Table 5.

B. PREPROCESSING

Images were parsed into N_{IDC} (No Cancer) and P_{IDC} (IDC(+)) lists, resized to 50 × 50 pixels using OpenCV and normalized to [0, 1]. Current diagnostic methods suffer from pathologist shortages, subjective interpretations and heavy workloads, necessitating automated deep-learning solutions like HistoDX. Two imbalance correction methods were applied:

Oversampling: Duplicated P_{IDC} samples to balance the training set.

Weighted Loss: A weighted categorical cross-entropy loss assigned higher weights to IDC(+) (weight = 2.5).

Weighted loss and oversampling mitigated imbalance effectively (97% accuracy), though advanced methods like GANs are planned for future robustness. To enhance robustness, we applied minimal augmentation: `RandomRotation(0.0139)` for ±5° rotations to mimic tissue orientation variance and `RandomContrast(0.1)` to adjust for staining differences, preserving histopathological integrity over aggressive methods like flipping that could distort diagnostic features.

The dataset, comprising 277,524 histopathology images from the IDC dataset, was first shuffled to ensure randomness and then split into 80% training (222,019 images) and 20% testing (55,505 images) subsets using `train_test_split` from `sklearn.model_selection`. Labels were one-hot encoded via Keras' `to_categorical` to facilitate multi-class classification.

C. CUSTOMIZED MODEL EfficientNetV2-B3

In this study, EfficientNetV2-B3 was adopted as the core architecture for breast cancer detection using histopathological images, owing to its robust theoretical underpinnings and practical advantages tailored to the complexities of this data type. EfficientNetV2 employs a compound scaling strategy that optimizes network depth, width and resolution, achieving high performance with reduced computational overhead [45]. The B3 variant was selected for its balanced capacity, offering sufficient power to process large-scale histopathological datasets while remaining computationally feasible within the resource constraints of this study.

TABLE 2. Characteristics of IDC Histopathology, BreakHis, and BACH datasets used in HistoDX evaluation.

Dataset	Source	Image Size (pixels)	Magnification	Class Distribution	Total Images
IDC Histopathology (Paul Mooney)	Kaggle, 279 patients	50×50 (preprocessed)	40x	28.4% IDC(+), 71.6% No Cancer	277,524
BreakHis	P&D Lab, Brazil	224×224 (resized)	40x	Benign (25.1%), Malignant (74.9%)	7,909
BACH	ICIAR 2018 Challenge	300×300 (resized)	20x, 40x (mixed)	Normal (25%), Benign (25%), In Situ (25%), Invasive (25%)	400

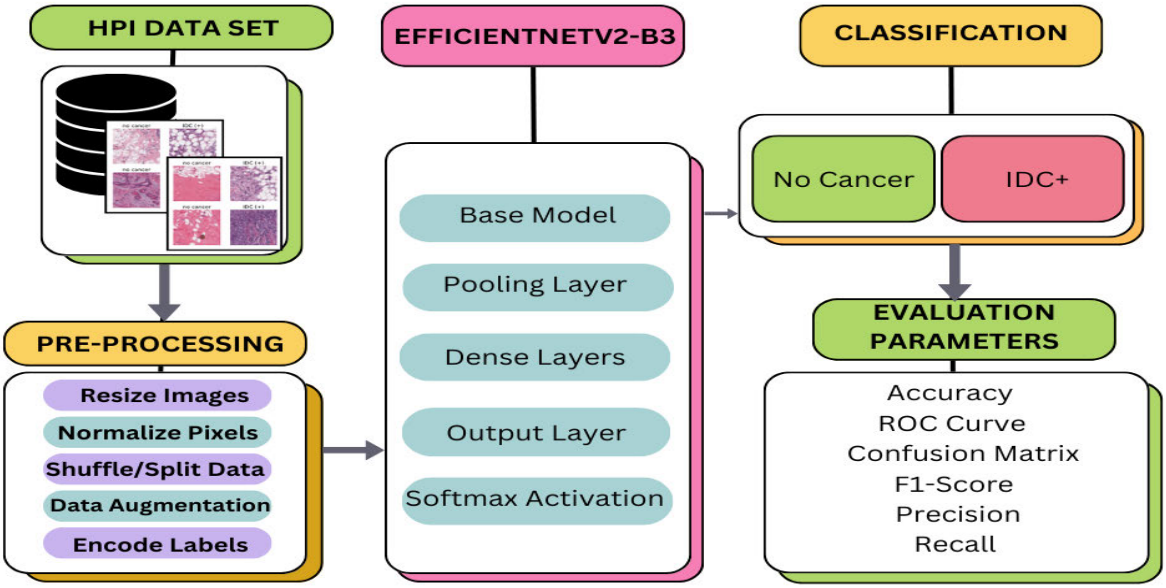


FIGURE 3. Sequence of operations performed by the proposed model.

Histopathological images present distinct challenges for breast cancer detection, including the need to discern subtle cellular features such as nuclear pleomorphism, mitotic figures and stromal interactions within highly variable tissue samples. EfficientNetV2-B3 addresses these demands through its advanced feature extraction capabilities, enabled by Fused-MBConv blocks and improved training efficiency, which enhances its ability to model fine-grained patterns compared to traditional architectures like ResNet or DenseNet [45]. In preliminary experiments, ResNet-50 and DenseNet-121 were tested but proved less optimal: ResNet-50 demanded significantly higher computational resources, while DenseNet-121 struggled to achieve comparable accuracy in detecting cellular-level details critical for histopathological analysis, particularly under the memory limitations of Google Colab. By contrast, EfficientNetV2-B3 demonstrated superior performance in pilot tests, achieving higher accuracy with fewer parameters, making it an ideal choice for this task. Moreover, the model's efficiency supports its potential for practical deployment in clinical settings, where rapid and precise analysis of histopathological slides is paramount. Leveraging transfer learning with ImageNet pre-trained weights, fine-tuned on our histopathological dataset, EfficientNetV2-B3 effectively adapted to domain-specific features, enhancing its discriminative power for breast cancer classification. This choice reflects a deliberate integration of theoretical rigour, empirical efficacy and practical feasibility, positioning EfficientNetV2-B3 as uniquely suited to address the challenges of breast cancer detection in histopathological imaging.

Compound scaling is employed by EfficientNetV2-B3, a scaled-up version of EfficientNet, to balance resolution, depth and width. While EfficientNetV2 focuses on improving training speed and accuracy, the original EfficientNet established scaling ideas to optimize network performance. Developments such as Fused MBConv layers, which integrate depthwise separable convolutions and residual connections to reduce the training period without sacrificing performance, are responsible for this gain. Furthermore, the approach incorporates sophisticated regularization and augmentation algorithms in addition to optimizations for quicker training. Due to these improvements, EfficientNetV2-B3 combines outstanding accuracy and efficiency, which makes it appropriate for a variety of applications.

As EfficientNetV2-B3 has been viewed in Fig. 3 as a modern pre-trained model with outstanding accuracy in image classification tasks, it was chosen as the basis for the customized convolutional neural network (CNN) model. Due to its perfect mix of accuracy and computational economy, especially in image classification uses, EfficientNetV2-B3 was chosen over other models in the EfficientNet family. It is a practical feature extractor due to its pre-trained

weights on ImageNet, which is essential when working with smaller datasets like histopathology images. Due to its ability to balance model size and accuracy, EfficientNetV2-B3 is renowned for its excellent adaptability and high performance, making it ideal for fine-tuning domain-specific datasets.

EfficientNetV2-B3 was selected over alternatives like ResNet or earlier EfficientNet variants due to its compound scaling, which optimizes depth, width and resolution and its training-aware optimizations (e.g., Fused MBConv), enhancing feature extraction for subtle histopathological patterns critical in breast cancer diagnosis. This architecture delivers reliable and diverse feature extraction capabilities, having been pre-trained on the ImageNet dataset. The top layers of EfficientNetV2-B3 were omitted by setting `include_top=False` in order to customize the model for the particular purpose of categorizing histopathological image patches. This made room for the addition of custom classification layers to address the problem at hand while maintaining the use of its feature extraction layers.

The addition of more layers and hyperparameters enhances the model's ability to carry out classification tasks effectively. To increase the model's capability and allow it to identify increasingly complex patterns in the data, dense layers are added. For instance, the non-linearity needed to capture complex interactions while minimizing the potential of vanishing gradients is provided by the addition of a fully linked layer with ReLU activation. Dropout layers, which are particularly helpful for smaller datasets, assist in preventing overfitting by randomly deactivating a small number of neurons during training.

To ensure interpretability for medical professionals, the model relies on its high accuracy (97%) and clear probabilistic outputs via Softmax, which indicate confidence in IDC(+) vs. No Cancer classifications, facilitating clinical decision-making; future work will explore visualization techniques like Grad-CAM. Hyperparameters are crucial for maximizing the model's performance. The optimizer's step size is affected by the learning rate, which achieves a balance between training stability and convergence speed. An adequately considered learning rate guarantees effective gradient descent without overshooting minima. Each robust layer's capability is measured by the number of neurons within it, which needs to be adjusted based on how complex the task is. Last but not least, activation functions add crucial non-linearity. Softmax in the output layer converts logits into class probabilities, making it ideal for binary or multi-class classification problems, while ReLU effectively handles gradient flow in intermediate layers.

Several custom layers were added to the EfficientNetV2-B3 in Fig. 4 backbone in order to change it for binary classification (IDC(+) vs No Cancer).

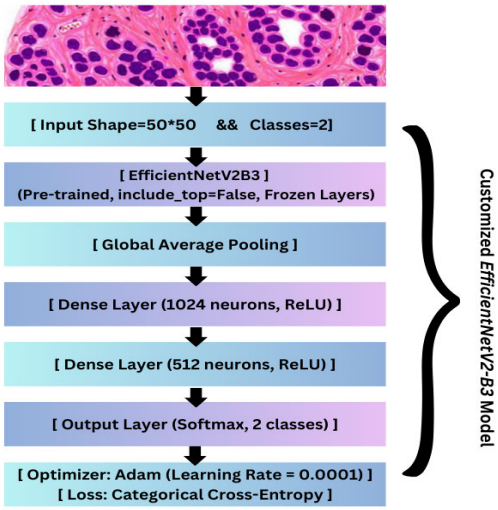


FIGURE 4. Customized layers of the proposed model.

1) INPUT LAYER

The model begins with a $50 \times 50 \times 3$ input image that includes three color channels that indicate the red, green and blue (RGB) intensities at each pixel. The image is 50 pixels broad and 50 pixels high. Every image in the context of breast cancer detection depicts a patch of histopathology slides containing detailed information regarding tissue structures. The model's three color channels enable it to capture small characteristics and color fluctuations in the tissue, including aberrant cell development and staining patterns, which are two critical indicators of malignant areas. Through the model's evaluation of these images, the network develops the ability to identify intricate patterns in the tissue, which helps it distinguish between IDC(+) and No Cancer cells. Consequently, it aids in the early and precise detection of cancer.

2) BASE MODEL (HistoDX)

The HistoDX model, built upon the EfficientNetV2B3 base architecture, which was previously trained on ImageNet, is then applied to the input image. The model is already able to identify significant data from a variety of images like shapes, edges and textures. These pre-trained features are essential for cancer detection because they save the model from having to learn essential low-level features from scratch, enabling it to focus on pertinent patterns in the tissue images. Only the convolutional layers remain when the final classification layers are eliminated by using the EfficientNetV2B3 model with the `include_top=False` parameter. This is important because feature extraction, which is the main task of finding pertinent patterns in the input data, is handled by the convolutional layers. The model can extract information from the histopathological images of IDC(+) and No Cancer tissue, which can often be fragile and complex due to the pre-trained weights in these layers.

$$X_{\text{base}} = \text{EfficientNetV2B3}(X_{\text{input}}) \tag{1}$$

Employing previously trained models, such as EfficientNetV2B3, makes the most of transfer learning. The model can

TABLE 3. Customized model details.

Aspect	Values
Backbone Model	EfficientNetV2-B3 (pre-trained on ImageNet)
Input Shape	(50, 50, 3)
Freezing Layers	All layers frozen (`trainable=False`)
Custom Layers	GAP, Dense (1024, ReLU), Dense (512, ReLU), Output (2, Softmax)
Loss Function	Categorical Cross-Entropy
Optimizer	Adam (learning rate: 0.0001)
Metric	Accuracy
Training Configuration	50 epochs, batch size 64, validation split
Model Purpose	IDC(+)/No Cancer classification

use the knowledge acquired from ImageNet to extract high-level features, which may be modified for the specific task of cancer evaluation, rather than training a model from scratch, which would require a significant amount of information and computational resources. This significantly improves performance, mainly when the dataset is smaller or includes highly specialized data, such as histopathology images.

3) GLOBAL AVERAGE POOLING (GAP)

Global Average Pooling (GAP), which decreases the spatial dimensions of the feature maps yet preserves significant high-level features, is crucial when employing EfficientNetV2B3 for breast cancer diagnosis and following the extraction of feature maps from the input image by the base model. GAP reduces the data to a single scalar value per channel by computing the average value for each feature map (channel). Through doing this, the model is able to ignore less significant spatial features like background noise and focus on pertinent patterns, such as abnormal tissue structure or cancerous growths. GAP helps the model to generalize better and increase classification accuracy by condensing the most critical data, particularly in complex histopathology images. The model's effectiveness is improved by this reduction in complexity, which also lowers the likelihood of overfitting and enhances the model's ability to identify malignant tissues.

$$Y_k = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X[i, j, k] \tag{2}$$

where H and W are the height and width of the feature map, respectively and k refers to the channel.

4) DENSE LAYER 1 (1024 NEURONS)

The output from the global average pooling is passed through a dense layer with 1024 neurons and ReLU activation. This dense layer helps the model learn more complex patterns from the extracted features. With 1024 neurons and ReLU activation, the dense layer helps the model learn complex patterns from the information that the basic model retrieved and that Global Average Pooling processed. This layer's increased neuron count enables the model to recognize more complicated associations in the data, which is crucial in finding minute patterns in tissue that might indicate signs of malignancy. By adding non-linearity, the ReLU activation enhances the model's ability to differentiate between IDC(+) and No Cancer tissues, improving classification accuracy.

$$Y_1 = \text{ReLU}(W_1 \cdot X + b_1) \tag{3}$$

where W_1 is the weight matrix, X is the input from the GAP layer and b_1 is the bias term. The ReLU activation function ensures that the output is non-negative.

5) DENSE LAYER 2 (512 NEURONS)

The patterns acquired from the first dense layer are further enhanced by the second dense layer, which has 512 neurons and ReLU activation. This second layer allows the model to process the data in more detail and develop finer, more precise representations of the data after the first dense layer already identified the complex aspects. The model highlights the most vital characteristics by reducing the number of neurons from 1024 to 512, assisting in improving generalization and minimizing overfitting and accurately distinguishing between IDC(+) and No Cancer tissues. It requires the model's capacity to capture non-linear connections in the data, which is made possible by the ReLU activation. By concentrating on important patterns that suggest the existence of breast cancer, this additional component enhances the ability of the model to produce accurate predictions.

$$Y_2 = \text{ReLU}(W_2 \cdot Y_1 + b_2) \tag{4}$$

where W_2 is the weight matrix, Y_1 is the output from the first dense layer and b_2 is the bias term.

6) OUTPUT LAYER (SOFTMAX ACTIVATION)

The final dense layer, containing two neurons and representing the two classifications of IDC(+) and No Cancer, gets the output from the second dense layer. The softmax activation function in the final layer transforms the raw output values, also known as logits, into probabilities that add up to 1. It is easier to understand the predictions since the softmax function makes sure that the model's output accurately depicts the likelihood of each class. The neuron with the highest probability represents the anticipated class.

As an illustration, if the model generates a greater likelihood for the IDC(+) class, it indicates cancer is present. Still, a higher probability for the No Cancer class means that cancer is not present. For multi-class classification problems, this softmax activation is a necessity since it provides a straightforward choice between the possible outcomes:

$$Y_{\text{output}} = \text{Softmax}(W_3 \cdot Y_2 + b_3) \tag{5}$$

where W_3 is the weight matrix, Y_2 is the output from the second dense layer and b_3 is the bias term.

7) OPTIMIZER AND LOSS FUNCTION

The Adam optimizer, which modifies the learning rate depending on the gradient, is used to train the model using a learning rate of 0.0001 as shown in Table. 3. It guarantees stable and effective training. Since categorical cross-entropy evaluates the disparity between the actual labels and the projected probabilities, it is the perfect loss function for multi-class classification tasks. This combination enhances the model's capacity to correctly classify images as either IDC(+) or No Cancer by enabling it to learn by modifying weights to minimize the loss.

8) TRAINING CONFIGURATION

The model is trained using a batch size of 64 and for 50 epochs, with a validation split to evaluate performance on unseen data. In order to prevent overfitting and make sure the model performs efficiently when compared to newly collected breast cancer images, a validation split is used to assess the model's performance on unknown data. Hyperparameters were tuned by evaluating a range of learning rates (e.g., 0.0001, 0.001) and batch sizes (e.g., 16, 32, 64), selecting 0.0001 and 64 for optimal convergence and a test accuracy of 97% with 0.91 ROC-AUC.

9) EXPLAINABILITY ENHANCEMENT WITH SALIENCY MAPS

To enhance the interpretability of HistoDX and address the need for transparent decision-making in clinical settings, we implemented an enhanced saliency map technique to visualize the regions of histopathological images most influential to the model's predictions. Saliency maps highlight the areas of an input image that contribute most to the output class score by computing the gradient of the predicted class score with respect to the input image. Our implementation, built using TensorFlow, preprocesses images to a standardized 50 × 50 pixel resolution, matching the input shape of EfficientNetV2-B3 and normalizes pixel values to [0, 1].

The saliency map is generated by calculating the maximum absolute gradient across the RGB channels, followed by normalization and optional Gaussian smoothing (sigma = 1.5) to reduce noise and enhance visual clarity. This process is detailed in Algorithm 1, adapted for our binary classification task (IDC(+) vs No Cancer). The resulting saliency maps use a 'jet' color map, where red indicates strong influence and blue indicates low influence, providing pathologists with intuitive visual insights into the model's focus areas, such as abnormal cellular structures indicative of IDC. This approach bridges the gap between high accuracy and clinical trust by making the model's decision-making process more transparent.

Algorithm 1 Saliency Map Generation for Histopathology

Require: Model M, image X ($50 \times 50 \times 3$), class index k (optional), smoothing s, σ
Ensure: Saliency map S, class index k

```
Convert X to tensor T_X.
Get predictions P = M(T_X).
if k is not provided then
    k ← arg max P[0]
end if
Compute gradient G = ∇_{T_X} P[0, k].
S ← max(|G|, axis = −1).
Normalize S ← (S − min(S)) / (max(S) − min(S) + 10^−8).
if s is True then
    S ← GaussianFilter(S, σ)
end if
return S, k
```

IV. EXPERIMENTAL RESULTS

The models were implemented using the TensorFlow and Keras libraries, which provided high-level components across several tiers. Model with the best performance was selected for evaluation and compared with existing models in the field. This comparison involved training on the IDC dataset. The experiments were conducted in Google Colab, utilizing its free 12 GB RAM and Tesla K80 GPU to ensure computational efficiency, enabling rapid experimentation. As a result, the selected model achieved a test accuracy of 97% and ROC-AUC of 0.91, outperforming previous models on the IDC dataset Table 6. However, additional validation on other datasets is required to confirm its generalizability.

A. EVALUATION MATRICES

This section lists the evaluation metrics used to gauge the model's performance. These parameters include accuracy, ROC-AUC, confusion matrix and other statistics like precision, recall and F1-score. The results acquired in this study are explained and analyzed after the formulas for these parameters are given.

1) ACCURACY

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \tag{6}$$

The model has achieved an overall accuracy of 97%, which indicates the percentage of correctly classified samples. This result is calculated using the confusion matrix values, where FN = 1,515, TP = 959, TN = 52,638 and FP = 393. The accuracy formula results in the following:

$$\text{Accuracy} = \frac{52638 + 959}{52638 + 959 + 1515 + 393} = 97\%$$

This high accuracy is beneficial in distinguishing between IDC(+) and No Cancer samples. As demonstrated by Fig. 7, our customized model has a 97% testing accuracy and a 98% training accuracy. These high accuracy values indicate the extent to which the model generalizes to new data and how well it learned from the training set. A well-balanced performance is shown by a slight variation in training and testing accuracy, which points to minimal overfitting and strong learning. Effective handling of data is the model's strongest suit and this is made possible by a carefully planned architecture, ideal hyperparameter tuning and an adequate split of training and testing data. This 97% accuracy, though high, reflects robust tuning (learning rate 0.0001, batch size 64) and effective class imbalance mitigation via oversampling and weighted loss, despite challenges in detecting the under sampled IDC(+) class (FNR 61.2%), as detailed below. When applied to real-life scenarios, this balance ensures the model's ability to forecast and reliability, showing its adaptability and resilience.

2) PRECISION,RECALL AND F1-SCORE

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

Precision, recall and F1-score provide a deeper understanding of the model's performance for both IDC(+) and No Cancer samples. For No Cancer samples, precision is 0.97, recall is 0.99 and F1-score is 0.98, indicating exceptional performance. For IDC(+) samples, precision is 0.71, recall is 0.39 and F1-score is 0.50. This reflects the challenges in accurately detecting this under sampled class, as shown in Table. 4.

The macro average yields a precision of 0.84, recall of 0.69 and F1-score of 0.74. It highlights the room for improvement in balancing the performance of both classes. The weighted average, considering the support of each class, provides precision, recall and F1-score of 0.96, 0.97 and 0.96, respectively, showing strong overall performance in Fig. 5. The lower IDC(+) recall (0.39) stems from class imbalance (28.4% IDC(+) vs. 71.6% No Cancer), addressed with basic oversampling and weighted loss (IDC(+) weight = 2.5), though advanced methods like GANs could further enhance under sampled class detection.

3) ROC-AUC (AREA UNDER CURVE)

ROC-AUC measures the ability of the model to distinguish between classes across various thresholds. Plotting the true positive rate (TPR) against the false positive rate (FPR) across various thresholds accommodates the ROC curve to be used to assess the model's performance. The performance of the overall model is represented by the area Under the Curve (AUC). The average area under the curve score for our customized model is 0.91%, as shown in Fig. 6, which signifies excellent discrimination power. An AUC score closer to 1 indicates an effective classifier. This 0.91 ROC-AUC supports the model's strong class separation, validated by empirical hyperparameter tuning, though its clinical utility depends on improving IDC(+) sensitivity.

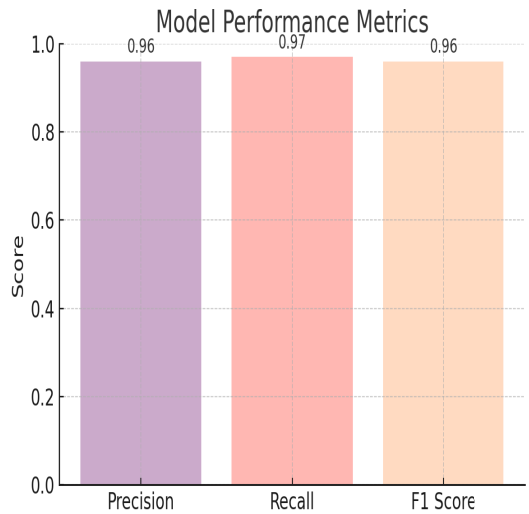


FIGURE 5. Comparison of precision, recall and F1-Score.

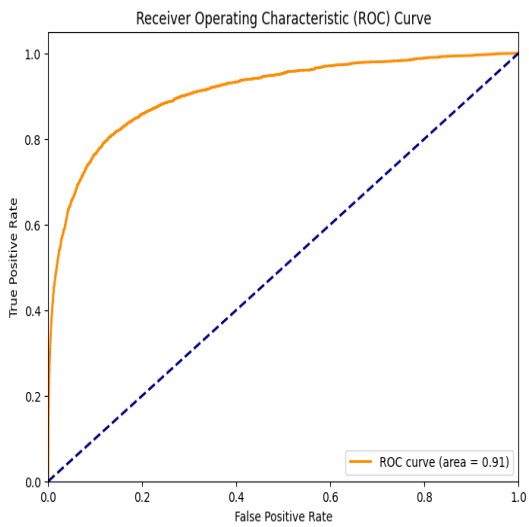


FIGURE 6. ROC score of proposed model.

4) LOSS

Loss quantifies the error between predicted and actual labels and is calculated using the categorical cross-entropy function for multi-class classification:

$$\text{Loss} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} y_{ij} \log(p_{ij}) \tag{10}$$

where y_{ij} is the true label and p_{ij} is the predicted probability for class j of sample i. As can be seen in Fig. 7, our customized model showed a testing loss of 0.12 and a training

TABLE 4. Detailed performance metrics.

Metric	Precision	Recall	F1-Score	Specificity	NPV	FNR	FPR	MCR
IDC(+)	0.71	0.39	0.50	0.99	0.97	0.61	0.01	0.03
No Cancer	0.97	0.99	0.98	0.99	0.97	0.01	0.01	0.03
Macro Average	0.84	0.69	0.74	0.99	0.97	0.31	0.01	0.03
Weighted Average	0.96	0.97	0.96	0.99	0.97	0.31	0.01	0.03

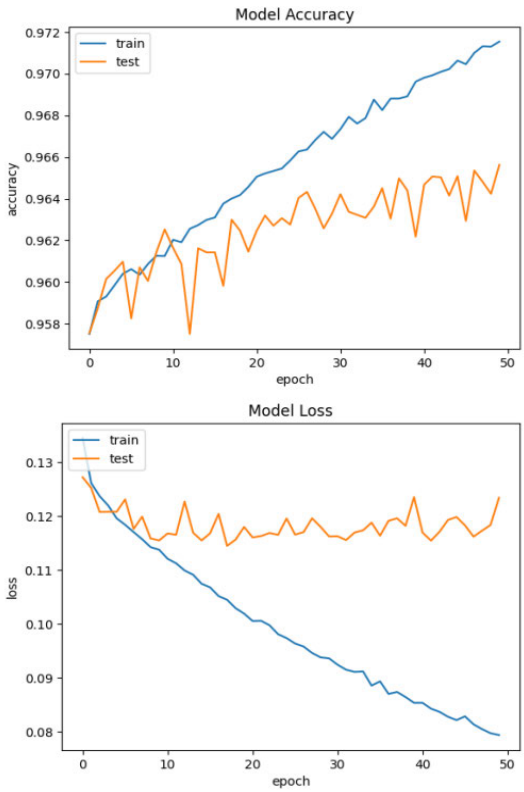


FIGURE 7. Accuracy and loss of proposed model.

loss of 0.08. These figures demonstrate that the model has successfully learned the fundamental patterns in the data, as proven by the minor mistakes it made during both the training and testing phases. The desirable characteristic of the model is that it performs somewhat better on unseen data, as indicated by the testing loss, which is slightly smaller than the training loss. Strong regularization strategies or a well-designed architecture that reduces overfitting which could be responsible for this behavior. The model's excellent generalization ability can be seen by the observed performance, which guarantees that it can adjust well to unique, unseen instances without sacrificing accuracy or reliance. This thorough training procedure demonstrates how well the customization and optimization techniques used throughout the model's development.

5) MISCLASSIFICATION RATE (MCR)

The Misclassification Rate (MCR) is the proportion of incorrectly classified instances (both false positives and false negatives). It is calculated as:

$$\text{MCR} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{11}$$

For our model, the MCR is calculated as:

$$\text{MCR} = \frac{393 + 1,515}{52,638 + 959 + 393 + 1,515} = 3\%$$

This result suggests that only 3% of the samples are misclassified, indicating good overall performance.

6) SPECIFICITY

Specificity measures the proportion of actual negatives (No Cancer samples) that are correctly identified. It is calculated as:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{12}$$

For our model, Specificity is calculated as:

$$\text{Specificity} = \frac{52,638}{52,638 + 393} \approx 0.9926\,(99.26\%)$$

This indicates that the model is correctly identifying 99.26% of the No Cancer samples.

7) FALSE NEGATIVE RATE (FNR)

The False Negative Rate (FNR) measures the proportion of actual positives IDC(+) that are incorrectly predicted as negative (No Cancer). It is calculated as:

$$\text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}} \tag{13}$$

For our model, FNR is calculated as:

$$\text{FNR} = \frac{1,515}{1,515 + 959} \approx 0.613\,(61.3\%)$$

This relatively high FNR suggests that the model misses a significant proportion of IDC(+) samples. It underscores the challenge of class imbalance, mitigated partially by oversampling and weighted loss, with potential clinical implications for missed diagnoses.

8) FALSE POSITIVE RATE (FPR)

The False Positive Rate (FPR) measures the proportion of actual negatives (No Cancer) that are incorrectly predicted as positives IDC(+). It is calculated as:

$$\text{FPR} = \frac{\text{FP}}{\text{TN} + \text{FP}} \tag{14}$$

For our model, FPR is calculated as:

$$\text{FPR} = \frac{393}{52,638 + 393} \approx 0.0074\,(0.74\%)$$

This FPR suggests that the model is making a small number of false positive predictions for No Cancer samples.

9) NEGATIVE PREDICTIVE VALUE (NPV)

The Negative Predictive Value (NPV) measures the proportion of negative predictions (No Cancer) that are actually correct. It is calculated as:

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \tag{15}$$

For our model, NPV is calculated as:

$$\text{NPV} = \frac{52,638}{52,638 + 1,515} \approx 0.972\,(97.2\%)$$

This NPV indicates that 97.2% of the samples predicted as No Cancer are actually correct.

10) CONFUSION MATRIX

The confusion matrix gives a thorough analysis of the model's performance. It demonstrates that the model accurately categorizes IDC(+) samples (959) as true positives and No Cancer samples (52,638) as true negatives. However, as illustrated in Fig. 8, IDC(+) samples (1,515) were incorrectly identified as No Cancer (false negatives) and No Cancer samples (393) were incorrectly classified as IDC(+) (false positives).

These findings reveal that while the model excels at identifying samples without cancer, it struggles to detect all IDC(+) cases. This limitation is primarily due to the dataset's class imbalance, where the under sampled class IDC(+) has fewer samples, making it harder for the model to generalize effectively across this class. To aid medical professionals, the confusion matrix and Softmax probabilities provide interpretable insights into classification confidence, though future visualization tools could enhance trust. The false positive rate (FPR) highlights this challenge further. FPR measures the proportion of actual No Cancer samples that are incorrectly classified as IDC(+). In this model, the FPR is calculated as:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} = \frac{393}{393 + 52,638} \approx 0.0074\,(0.74\%).$$

This indicates that a small proportion of No Cancer samples are being flagged as IDC(+). Although this FPR appears low, in large-scale screening scenarios, even a small percentage of false positives can result in a substantial number of misclassifications, leading to unnecessary stress for patients and additional diagnostic procedures that could burden healthcare systems. A high false negative rate (FNR) also highlights the importance of minimizing missed detections of IDC(+):

$$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}} = \frac{1,515}{1,515 + 959} \approx 0.613\,(61.3\%).$$

This relatively high FNR suggests that nearly half of IDC(+) samples are missed, potentially delaying treatment for affected individuals. The combination of a low FPR and a high FNR underscores the trade-off between avoiding false positives and minimizing missed IDC(+) cases. This trade-off suggests clinical potential for screening but highlights the need for improved IDC(+) detection to reduce missed cases, possibly via advanced techniques or real-time integration with pathology workflows. Addressing this trade-off will require strategies such as balancing the dataset, fine-tuning the decision threshold or using advanced techniques like cost-sensitive learning to optimize the model for both precision and recall in detecting IDC(+).

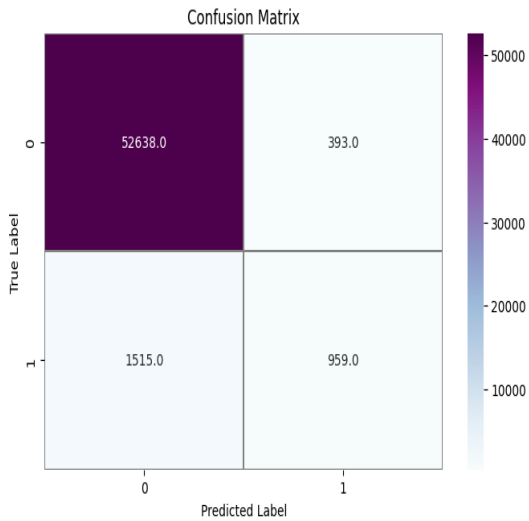


FIGURE 8. Confusion matrix score of proposed model.

B. VALIDATION ACROSS MULTIPLE DATASETS

To evaluate the robustness of the HistoDX model, we tested its performance on the BreakHis dataset and BACH dataset. The dataset differs in source, preprocessing and characteristics, providing a comprehensive assessment of the model's generalizability. The images used were 224 × 224 pixels in size and were sourced from the BreakHis dataset, which consists of images captured at 40x magnification. The performance metrics, including training accuracy, training loss, validation accuracy, validation loss, overall accuracy, precision, recall and F1-score, are presented in Table 5. The evaluation results of the HistoDX model on the BreakHis dataset demonstrate its high performance and robustness. Achieving an overall accuracy of 97%, the model shows strong effectiveness in classifying both benign and malignant breast cancer images. The weighted average results for the HistoDX model highlight its overall balanced performance across both benign and malignant categories, taking into account the class distribution in the dataset. The weighted average precision, recall and F1-score are all 0.97, demonstrating that the model performs consistently

TABLE 5. Performance metrics of HistoDX Across BreakHis, BACH, and IDC Histopathology datasets.

Dataset	Train Acc.	Train Loss	Val Acc.	Val Loss	Overall Acc.	Precision	Recall	F1-Score
BreakHis	0.99	0.01	0.97	0.08	0.97	0.97	0.97	0.97
BACH	0.99	0.02	0.90	0.42	0.90	0.90	0.90	0.90
IDC Histopathology Dataset by Paul Mooney	0.97	0.08	0.97	0.12	0.97	0.96	0.97	0.96

well while accounting for the imbalance between the two classes. The high weighted average precision indicates that, on average, the model is highly accurate when classifying both benign and malignant cases. Similarly, the weighted average recall of 0.97 shows that the model is effective at identifying both classes without favoring one over the other. The weighted average F1-score, also 0.97, further reinforces this balanced performance, suggesting that the model maintains an effective trade-off between precision and recall, which is essential for real-world applications where both types of errors (false positives and false negatives) need to be minimized. Overall, the weighted average results reflect the model's robustness and generalizability across the dataset, making it a reliable tool for classifying breast cancer images in a clinical setting.

To assess the robustness of the HistoDX model, we evaluated its performance on the BACH dataset. This dataset, distinct in source, preprocessing, and characteristics, provides a thorough test of the model's generalizability. The images were resized to 300×300 pixels and sourced from the BACH dataset, which contains histopathology images with varied magnifications (20x and 40x) and staining protocols. The evaluation results of the HistoDX model on the BACH dataset underscore its performance in classifying breast cancer images, though with slightly lower metrics compared to the BreakHis dataset. As shown in Table 5, model has achieved an overall accuracy of 90% on BACH dataset, the model demonstrates reasonable effectiveness in distinguishing between benign and malignant cases. The weighted average results for the HistoDX model on the BACH dataset show a balanced performance across both categories, accounting for the class distribution. The weighted average precision, recall, and F1-score are all 0.90, indicating consistent performance despite the dataset's unique characteristics, such as differences in source and preprocessing. The weighted average precision of 0.90 suggests that the model maintains a good level of accuracy in its classifications, while the weighted average recall of 0.90 reflects its ability to identify both benign and malignant cases without significant bias toward either class. The weighted average F1-score of 0.90 further confirms the model's ability to balance precision and recall, which is critical for clinical applications where minimizing both false positives and false negatives is important. Overall, the results on the BACH dataset highlight the HistoDX model's adaptability, though its performance indicates room for improvement in generalizing to more diverse datasets.

C. SALIENCY MAP VISUALIZATION RESULTS

To demonstrate the practical utility of our saliency map implementation, we applied it to a sample IDC(+) image from the IDC Histopathology Dataset by Paul Mooney (`10253_idx5_x501_y401_class1.png`). The image, preprocessed to 50×50 pixels due to computational constraints, was fed into HistoDX, and a saliency map was generated for the predicted IDC(+) class. As shown in Fig. 9, the original image (left) is paired with its corresponding saliency map (right), where color intensity, rendered using the 'jet' colormap, indicates the regions most influential to the model's decision. The map prominently highlights areas with abnormal cellular patterns, such as dense nuclear clustering and irregular cell structures, which align closely with histopathological features characteristic of invasive ductal carcinoma (IDC). The saliency map, with a shape of (1, 50, 50), was smoothed using a Gaussian filter (sigma = 1.5) to enhance interpretability by reducing high-frequency noise, making it easier for pathologists to discern critical regions.

This visualization confirms that HistoDX effectively focuses on diagnostically relevant regions, supporting its high performance of 97% accuracy and 0.91 ROC-AUC on the IDC dataset. The alignment between highlighted areas and IDC-specific features underscores the model's ability to prioritize clinically significant patterns, enhancing trust in its predictions. To further explore its interpretability, we qualitatively assessed saliency maps for additional IDC(+) and No Cancer samples, noting that No Cancer maps emphasized normal tissue architecture, though with less intensity variation. However, further analysis across a broader and more diverse sample set is needed to validate the consistency of these findings, particularly for No Cancer cases where subtle differences may be harder to capture. Expanding this approach to include other visualization techniques, such as Grad-CAM, could provide complementary insights, potentially revealing finer details about the model's decision-making process and strengthening its utility in clinical settings.

V. DISCUSSION

Developing a deep learning-based model for the identification of breast cancer using histopathology images was the primary goal of this study. Our proposed model, HistoDX, based on EfficientNetV2-B3, showed a fantastic 97% accuracy and a ROC-AUC score of 0.91, as indicated in Table 6. These results demonstrate how well the model differentiates between samples with IDC(+) and No Cancer.

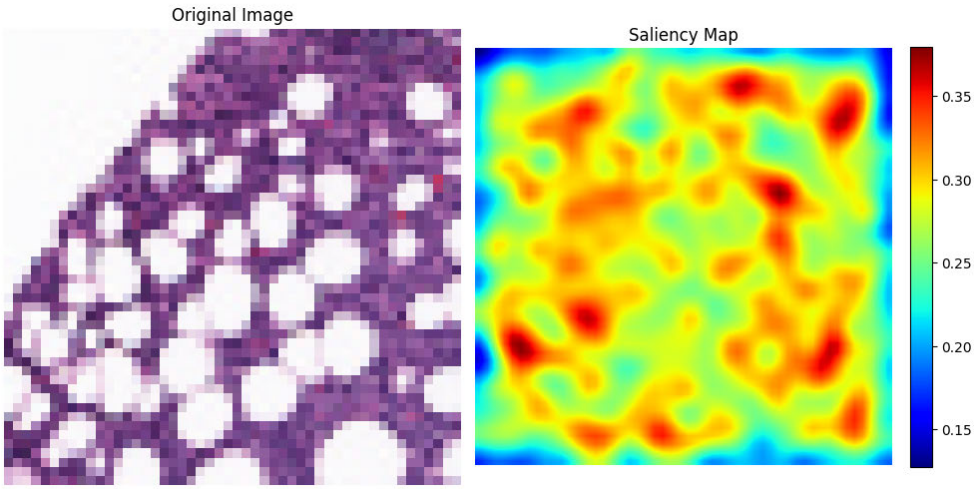


FIGURE 9. Saliency map visualization for an IDC(+) sample. Left: Original 50 × 50 histopathological image. Right: Corresponding saliency map highlighting influential regions (red = high influence, blue = low influence).

With a precision of 0.96 and recall of 0.97, the model's outstanding efficacy in detecting samples that lack cancer highlights its dependability in confidently classifying benign instances. This high performance addresses critical limitations in current diagnostic methods, such as pathologist shortages, subjective interpretations and heavy workloads, offering a scalable, automated alternative for breast cancer screening.

When compared to other recent studies in the field, where different methods have been tested on the IDC dataset, our model's 97% accuracy is a significant achievement. For example, Zhang et al. [35] reported an accuracy of 87.45% with a combination of MSRCNN and SVM, while Choudhary et al. [30] obtained an accuracy of 92.07% with a pruned ResNet50. With its 97% accuracy, our model beats other models such as Google Cloud AutoML (91.6%) by Zeng et al. [31] and EfficientNet (91.53%) by Souza et al. [38], both of which show good performance. The difficulties of achieving high performance in medical image processing are reflected in the notable accuracy of multiple additional models, such as Celik's ResNet-50 [32] and Abdolahi et al.'s baseline CNN model [33], both of which obtained 94.11% and 85%, correspondingly. The accuracy of 96.3% achieved by Zarif et al. [18], who combined CNN and EfficientNetV2-B3, is a very close comparison. The accuracy of our model, which also makes use of EfficientNetV2-B3, was somewhat higher at 97%. This implies that our model's higher performance was primarily due to the enhancements we made to it, including adjustments of hyperparameters, streamlining the architecture and utilizing strategies like transfer learning. EfficientNetV2-B3's compound scaling and training optimizations (e.g., Fused MBConv) provided superior feature extraction over alternatives like ResNet, while empirical tuning of learning rate (0.0001) and batch size (64) optimized convergence, yielding this 97% accuracy and 0.91 ROC-AUC, though IDC(+) sensitivity remains a challenge due to class imbalance.

Our model's great accuracy was mainly due to its customization. However, the state-of-the-art architecture of EfficientNetV2-B3 offers a strong basis because of its effectiveness and capacity to manage intricate image features. Our model's performance was further improved by carefully choosing the best preprocessing and training methods. Furthermore, we were able to improve generalization and produce more dependable outcomes by resolving problems like class imbalance and streamlining the model's learning process. Our method shows that a combination of well-designed model architecture and efficient fine-tuning is the key to superior performance when compared to other studies that used similar architectures, such as Melwin D. Souza's use of EfficientNet [38], which produced an accuracy of 91.53%. Furthermore, the success of our data preprocessing, which reduced the risk of overfitting while guaranteeing that both IDC(+) and No Cancer samples were fairly represented, is demonstrated by our model's capacity to attain high accuracy in spite of the class imbalance. However, the lower IDC(+) recall (0.39) reflects the persistent class imbalance (28.4% IDC(+) vs. 71.6% No Cancer), mitigated by basic oversampling and weighted loss (IDC(+) weight = 2.5). This suggests clinical potential for screening but limitations in diagnostic sensitivity.

To evaluate HistoDX's robustness beyond the IDC Histopathology Dataset by Paul Mooney, we applied the model to the BreakHis dataset and BACH dataset. Without any fine-tuning, HistoDX achieved an impressive overall accuracy of 0.97 on BreakHis and 0.90 on BACH, closely aligning with its performance on the IDC dataset (0.97). Remarkably, the result on BreakHis surpasses all previously reported benchmarks in the literature, including Zarif's 96% accuracy [18], while the 90% accuracy on

TABLE 6. Compilation of research studies on breast cancer detection.

Author Name (Reference)	Dataset	Model(s) Used	Accuracy (%)
Tejalal Choudhary [30]	IDC	ResNet50 (Pruned)	92.07%
Jianfei Zhang [35]	IDC	MSRCNN + SVM	87.45%
Melwin D Souza [38]	IDC	EfficientNet	91.53%
Yan Zeng [31]	IDC	Google Cloud AutoML	91.6%
Yusuf Celik [32]	IDC	ResNet-50	94.11%
Mohammad Abdolahi [33]	IDC	CNN (Baseline)	85%
Selina Sharmin [34]	IDC	ResNet50V2 + Light Boosting	95%
Nassib Abdallah [36]	IDC	Feature Engineering Pipeline	94.7%
Melwin D Souza [38]	IDC	EfficientNet	91.53%
Sania Eskandari [40]	IDC	ViT (Vision Transformer)	93%
Abhinav Kumar [41]	IDC	SLIDE-Net (EfficientNet + SequentialLSTM)	89%
M. Iffat Hossain [42]	IDC	Ensemble (ResNet-50)	94.05%
Sameh Zarif [18]	IDC	CNN + EfficientNetV2-B3	96.3%
Proposed Model	**IDC**	**HistoDX**	**97%**

BACH demonstrates solid generalization despite the dataset's diverse magnifications and preprocessing. These strong performances on BreakHis and BACH, without adaptation, highlight HistoDX's effectiveness in handling varied histopathological data distributions. These findings show that while the model excels on its original training dataset, it also achieves competitive results on new datasets like BreakHis and BACH, reinforcing its robustness and potential for further enhancement through techniques such as transfer learning or domain adaptation.

As demonstrated by the comparison with previous research, our model outperforms several modern approaches due to innovative customization, thoughtful architectural choices, and efficient training strategies. Achieving 97% accuracy on both the IDC Histopathology Dataset and the BreakHis dataset, alongside 90% accuracy on the BACH dataset, underscores the robustness and generalizability of our method. This strong performance, supported by the effective identification of breast cancer from histopathology images across diverse datasets, highlights the model's potential for practical applications in medical diagnostics. While initially trained on a single dataset, our multi-dataset evaluation reveals not only its strengths but also areas for improvement, particularly in optimizing performance on datasets like BACH with varied characteristics. For clinicians, the Softmax output provides interpretable confidence scores, adding trust and transparency. Future work could focus on enhancing performance through techniques such as addressing class imbalance, adapting to diverse magnifications, or expanding to whole-slide image analysis. Real-time clinical integration remains a promising avenue to maximize the model's impact.

A. LIMITATIONS

This study, however, has several shortcomings and limitations, even if the results are encouraging. A first could be a dataset class imbalance with significantly fewer IDC(+) samples, which will undoubtedly affect the model sensitivity towards this under sampled class, restraining its overall diagnostic efficacy. Furthermore, the examination of small image patches for classification may have resulted in a loss of surrounding contextual information important for accurate cancer detection, given that whole-slide images (WSIs) contain large amounts of beneficial spatial relationships. The model was trained and evaluated primarily on one dataset. We tested it on additional datasets which include BreakHis and BACH datasets. The model achieved equally strong performance on BreakHis (97% accuracy), demonstrating its generalizability across datasets despite the lack of fine-tuning but resizing images to 224×224 pixels due to computational limits may exclude essential tissue context. The BACH dataset, with 90% accuracy, highlights challenges from resizing images to 300×300 pixels, missing critical context, and handling mixed 20x/40x magnifications without optimization. Varied preprocessing and possible class imbalance may reduce sensitivity. While the IDC dataset spans 279 patients, implicitly capturing some variation in age and pathology, explicit diversity (e.g., race) is undocumented, further limiting generalizability. Furthermore, the

model's interpretability was partially addressed by integrating saliency maps, which help visualize decision-critical regions. However, interpretability remains a limitation, as more advanced and clinically meaningful methods such as Grad-CAM or attention-based explanations could provide deeper insights and improve clinical trust. Lastly, while effective for patch-based classification, the model does not address the challenge of whole-slide image analysis, which is a critical need in real-world applications.

B. FUTURE WORK

Future work will address the identified limitations to enhance HistoDX's performance across datasets. To tackle class imbalance, particularly with IDC(+) samples in the IDC dataset and potential imbalances in the BreakHis and BACH datasets, techniques like data balancing, cost-sensitive learning, or advanced augmentation using GANs could improve the detection of under sampled classes. Incorporating higher-level context by analyzing larger image portions or whole-slide images (WSIs) may boost diagnostic reliability, especially for BACH's complex dataset with mixed 20x/40x magnifications. HistoDX demonstrated strong generalizability, achieving 97% accuracy on both IDC and BreakHis datasets and 90% accuracy on BACH without fine-tuning. However, exploring transfer learning or domain adaptation could further optimize performance across diverse datasets like BACH, with its varied preprocessing and staining protocols. The current saliency map implementation supports explainability, but future efforts will integrate advanced visualization tools like Grad-CAM and analyze a broader, more diverse image set to clarify decision-making for pathologists. Generalizing the model to WSIs using weakly supervised learning and attention mechanisms will enable comprehensive analysis beyond isolated patches, particularly critical for BACH's heterogeneous images. Ensemble models could also be refined to combine the strengths of different approaches for improved accuracy and robustness. Ultimately, clinical validation studies are essential to assess HistoDX's real-world performance and impact on diagnostic workflows across IDC, BreakHis, and BACH datasets, paving the way for its development into a robust tool to augment breast cancer detection and support clinical decision-making.

VI. CONCLUSION

Using EfficientNetV2-B3 as the backbone architecture, we developed HistoDX, a deep learning-based model for identifying breast cancer from histopathology images. The model achieved an impressive 97% accuracy and a 0.91 ROC-AUC score on the IDC Histopathology Dataset by Paul Mooney, demonstrating its strong ability to distinguish between IDC(+) and No Cancer samples. It proved reliable for diagnosing No Cancer cases, with a precision of 0.96 and recall of 0.97. However, it was less effective at identifying IDC(+) samples, with a precision of 0.71, recall of 0.39, and F1-score of 0.50, likely due to the dataset's class imbalance (28.4% IDC(+) vs. 71.6% No Cancer), which hinders under sampled class detection despite mitigation efforts like over-sampling and weighted loss. Our results outperformed several recent studies, with the 97% accuracy surpassing approaches like Zarif's 96.3% [18] and Melwin D. Souza's 91.53% [38], driven by tailored modifications to EfficientNetV2-B3, hyperparameter tuning (learning rate 0.0001, batch size 64), and efficient preprocessing. To evaluate broader applicability, we applied HistoDX to the BreakHis Dataset (97% accuracy) and the BACH Dataset (90% accuracy) without fine-tuning, as shown in Table 5. These results exceed benchmarks for BreakHis and demonstrate competitive performance on BACH, despite its varied magnifications and preprocessing challenges. While IDC(+) detection remains a challenge, the model's high performance on No Cancer samples and its adaptability across IDC, BreakHis, and BACH datasets position it as a promising tool for breast cancer screening, addressing issues like subjectivity and workload in traditional diagnostics. However, clinical adoption requires improved IDC(+) sensitivity and enhanced interpretability using tools like saliency maps. Having evaluated HistoDX on multiple datasets, we observed consistent performance, with 97% accuracy on IDC and BreakHis and 90% on BACH, highlighting strong generalization without fine-tuning. Future research should focus on optimizing the model through transfer learning or domain adaptation, addressing class imbalance with methods like GANs or cost-sensitive learning, and extending to whole-slide image analysis to enhance real-world applicability across diverse datasets like BACH.

CODE AVAILABILITY

The source code used in this study is not publicly available due to ongoing development and institutional policies. It can be obtained upon reasonable request by contacting the corresponding author at shahzad.dar@superior.edu.pk.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude and appreciation to all those who have contributed to the successful completion of this research article.

REFERENCES

[1] C. Osborne, P. Wilson, and D. Tripathy, "Oncogenes and tumor suppressor genes in breast cancer: Potential diagnostic and therapeutic applications," *Oncologist*, vol. 9, no. 4, pp. 361–377, Jul. 2004.

[2] R. Yasin, G. Zafar, F. Rooman Ali Syed, S. Afzal, M. Fatima, Z. Rathore, A. Chughtai, and A. Chughtai, "CK5/6 expression in molecular subtypes of invasive ductal carcinoma," *Cureus*, vol. 16, no. 10, Oct. 2024, Art. no. e72608.

[3] P. Mishra, S. Biswas, and S. Dash, "Deep learning based biomedical named entity recognition systems," in *Deep Learning Techniques for Biomedical and Health Informatics*, vol. 68. Cham, Switzerland: Springer, 2019, pp. 23–40.

[4] T. G. Debelee, F. Schwenker, A. Ibenthal, and D. Yohannes, "Survey of deep learning in breast cancer image analysis," *Evolving Syst.*, vol. 11, no. 1, pp. 143–163, Mar. 2020.

[5] S. M. Abbas, N. Javaid, A. T. Azar, U. Qasim, Z. A. Khan, and S. Aslam, "Towards enhancing the robustness of scale-free IoT networks by an intelligent rewiring mechanism," *Sensors*, vol. 22, no. 7, p. 2658, Mar. 2022.

[6] T. Mahmood, A. Rehman, T. Saba, L. Nadeem, and S. A. O. Bahaj, "Recent advancements and future prospects in active deep learning for medical image segmentation and classification," *IEEE Access*, vol. 11, pp. 113623–113652, 2023.

[7] A. Kaur, C. Kaushal, J. K. Sandhu, R. Damaševičius, and N. Thakur, "Histopathological image diagnosis for breast cancer diagnosis based on deep mutual learning," *Diagnostics*, vol. 14, no. 1, p. 95, Dec. 2023.

[8] S. Iqbal, A. N. Qureshi, J. Li, and T. Mahmood, "On the analyses of medical images using traditional machine learning techniques and convolutional neural networks," *Arch. Comput. Methods Eng.*, vol. 30, no. 5, pp. 3173–3233, Apr. 2023.

[9] C. Mohanty, S. Mahapatra, B. Acharya, F. Kokkoras, V. C. Gerogiannis, I. Karamitsos, and A. Kanavos, "Using deep learning architectures for detection and classification of diabetic retinopathy," *Sensors*, vol. 23, no. 12, p. 5726, Jun. 2023.

[10] M. Chandrasekar, M. Ganesh, B. Saleena, and P. Balasubramanian, "Breast cancer histopathological image classification using EfficientNet architecture," in *Proc. IEEE Int. Conf. Technol., Eng., Manag. Societal Impact Using Marketing, Entrepreneurship Talent (TEMSMET)*, Dec. 2020, pp. 1–5.

[11] E. H. Houssein, M. M. Emam, A. A. Ali, and P. N. Suganthan, "Deep and machine learning techniques for medical imaging-based breast cancer: A comprehensive review," *Expert Syst. Appl.*, vol. 167, Apr. 2021, Art. no. 114161.

[12] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[13] Y. Guo, Y. Liu, A. Oerlemans, S. Lao, S. Wu, and M. S. Lew, "Deep learning for visual understanding: A review," *Neurocomputing*, vol. 187, pp. 27–48, Apr. 2016.

[14] L. Solorzano, S. Robertson, B. Acs, J. Hartman, and M. Rantalainen, "Ensemble-based deep learning improves detection of invasive breast cancer in routine histopathology images," *Modern Pathol.*, vol. 33, pp. 918–927, Jun. 2020.

[15] X. Wu, Y.-T. Zhang, K.-W. Lai, M.-Z. Yang, G.-L. Yang, and H.-H. Wang, "A fuzzy logic-integrated convolutional neural network approach for robust medical image analysis," *IEEE Trans. Fuzzy Syst.*, vol. 33, no. 1, pp. 94–107, Jan. 2025.

[16] M. A. Shoaib, K. W. Lai, J. H. Chuah, Y. C. Hum, R. Ali, S. Dhanalakshmi, H. Wang, and X. Wu, "Comparative studies of deep learning segmentation models for left ventricle segmentation," *Frontiers Public Health*, vol. 10, Aug. 2022, Art. no. 981019.

[17] P. S. Q. Yeoh, K. W. Lai, S. L. Goh, K. Hasikin, X. Wu, and P. Li, "Transfer learning-assisted 3D deep learning models for knee osteoarthritis detection: Data from the osteoarthritis initiative," *Frontiers Bioeng. Biotechnol.*, vol. 11, Apr. 2023, Art. no. 1164655.

[18] S. Zarif, H. Abdulkader, I. Elaraby, A. Alharbi, W. S. Elkilani, and P. Pławiak, "Using hybrid pre-trained models for breast cancer detection," *PLoS ONE*, vol. 19, no. 1, Jan. 2024, Art. no. e0296912.

[19] J. Potsangbam and S. S. Devi, "Classification of breast cancer histopathological images using transfer learning with DenseNet121," in *Proc. Int. Conf. Mach. Learn. Data Eng. (ICMLDE)*, vol. 235. Amsterdam, The Netherlands: Elsevier, Jan. 2024, pp. 1990–1997.

[20] S. Kumbhare, A. B. Kathole, and S. Shinde, "Federated learning aided breast cancer detection with intelligent heuristic-based deep learning framework," *Biomed. Signal Process. Control*, vol. 86, Sep. 2024, Art. no. 105080.

[21] X. Yi, E. Walia, and P. Babyn, "Applications of generative adversarial networks in medical imaging: A review," *J. Med. Imag. Radiat. Sci.*, vol. 58, Dec. 2019, Art. no. 101552.

[22] S. Singh, S. S. Rawat, M. Gupta, B. K. Tripathi, F. Alanzi, A. Majumdar, P. Khuwuthyakorn, and O. Thinnukool, "Hybrid models for breast cancer detection via transfer learning technique," *Comput., Mater. Continua*, vol. 74, no. 2, pp. 3063–3083, 2023.

[23] F. M. Talaat, S. A. Gamel, R. M. El-Balka, M. Shehata, and H. Zaineldin, "Grad-CAM enabled breast cancer classification with a 3D inception-ResNet v2: Empowering radiologists with explainable insights," *Cancers*, vol. 16, no. 21, p. 3668, Oct. 2024.

[24] M. J. Hasan, W. S. H. M. W. Ahmad, M. F. A. Fauzi, J. T. H. Lee, S. Y. Khor, L. M. Looi, F. S. Abas, A. Adam, and E. W. L. Chan, "Real-time segmentation and classification of whole-slide images for tumor biomarker scoring," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 36, no. 7, Nov. 2024, Art. no. 102204.

[25] M. M. Ali, F. Maqsood, S. Liu, W. Hou, L. Zhang, and Z. Wang, "Enhancing breast cancer diagnosis with channel-wise attention mechanisms in deep learning," *Comput., Mater. Continua*, vol. 77, no. 3, pp. 2699–2714, 2023.

[26] K. Gupta and N. Chawla, "Analysis of histopathological images for prediction of breast cancer using traditional classifiers with pre-trained CNN," *Proc. Comput. Sci.*, vol. 167, pp. 878–889, May 2020.

[27] X. Li, X. Shen, Y. Zhou, X. Wang, and T.-Q. Li, "Classification of breast cancer histopathological images using interleaved DenseNet with SENet (IDSNet)," *PLoS ONE*, vol. 15, no. 5, May 2020, Art. no. e0232127.

[28] X. Wang, I. Ahmad, D. Javeed, S. A. Zaidi, F. M. Alotaibi, M. E. Ghoneim, Y. I. Daradkeh, J. Asghar, and E. T. Eldin, "Intelligent hybrid deep learning model for breast cancer detection," *J. Healthcare Eng.*, vol. 2024, Jun. 2024, Art. no. 123456.

[29] N. A. Barsha, A. Rahman, and M. R. C. Mahdy, "Automated detection and grading of invasive ductal carcinoma breast cancer using ensemble of deep learning models," *Comput. Biol. Med.*, vol. 139, Dec. 2021, Art. no. 104931.

[30] T. Choudhary, V. Mishra, A. Goswami, and J. Sarangapani, "A transfer learning with structured filter pruning approach for improved breast cancer classification on point-of-care devices," *Comput. Biol. Med.*, vol. 134, Jul. 2021, Art. no. 104432.

[31] Y. Zeng and J. Zhang, "A machine learning model for detecting invasive ductal carcinoma with Google cloud AutoML vision," *Comput. Biol. Med.*, vol. 122, Jul. 2020, Art. no. 103861.

[32] Y. Celik, M. Talo, O. Yildirim, M. Karabatak, and U. R. Acharya, "Automated invasive ductal carcinoma detection based using deep transfer learning with whole-slide images," *Pattern Recognit. Lett.*, vol. 133, pp. 232–239, May 2020.

[33] M. Abdolahi, M. Salehi, I. Shokatian, and R. Reiazi, "Artificial intelligence in automatic classification of invasive ductal carcinoma breast cancer in digital pathology images," *Med. J. Islamic Republic Iran*, vol. 34, p. 140, Oct. 2020.

[34] S. Sharmin, T. Ahammad, M. A. Talukder, and P. Ghose, "A hybrid dependable deep feature extraction and ensemble-based machine learning approach for breast cancer detection," *IEEE Access*, vol. 11, pp. 87694–87708, 2023.

[35] J. Zhang, X. Guo, B. Wang, and W. Cui, "Automatic detection of invasive ductal carcinoma based on the fusion of multi-scale residual convolutional neural network and SVM," *IEEE Access*, vol. 9, pp. 32959–32971, 2021.

[36] N. Abdallah, J.-M. Marion, C. Tauber, T. Carlier, M. Hatt, and P. Chauvet, "Enhancing histopathological image classification of invasive ductal carcinoma using hybrid harmonization techniques," *Sci. Rep.*, vol. 13, no. 1, Nov. 2023, Art. no. 20014.

[37] A. Kajala and S. Jaiswal, *Classification of Breast Cancer Histopathology Images Using EfficientNet Architectures*. Cham, Switzerland: Springer, 2022, pp. 625–634.

[38] M. D. Souza, G. A. Prabhu, V. Kumara, and K. M. Chaithra, "EarlyNet: A novel transfer learning approach with VGG11 and EfficientNet for early-stage breast cancer detection," *Int. J. Syst. Assurance Eng. Manage.*, vol. 15, no. 8, pp. 4018–4031, Aug. 2024.

[39] S. Krishna, S. S. Suganthi, S. Krishnamoorthy, and A. Bhavsar, "Stain normalized breast histopathology image recognition using convolutional neural networks for cancer detection," 2022, *arXiv:2201.00957*.

[40] S. Eskandari, A. Eslamian, N. Munia, A. Alqarni, and Q. Cheng, "Evaluating deep learning models for breast cancer classification: A comparative study," 2024, *arXiv:2408.16859*.

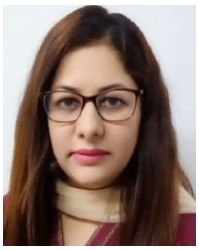
[41] A. Kumar, H. Tiwari, R. Singh, A. K. Singh, and S. K. Singh, "SLIDE-net: A sequential modeling approach with adaptive fuzzy C-mean empowered data balancing policy for IDC detection," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 10, pp. 5557–5570, Oct. 2024.

[42] M. I. Hossain, A. Billa, and M. E. Karim, "Prediction of IDC breast cancer by the application of transfer learning with an ensemble method," in *Proc. Int. Conf. Inf. Commun. Technol. Sustain. Develop. (ICICT4SD)*, Sep. 2023, pp. 239–243.

[43] M. Tafavvoghi, A. Sildnes, M. Rakaee, N. Shvetsov, L. A. Bongo, L.-T.-R. Busund, and K. Møllersen, "Deep learning-based classification of breast cancer molecular subtypes from H&E whole-slide images," *J. Pathol. Informat.*, vol. 16, Jan. 2025, Art. no. 100410.

[44] P. T. Mooney, "Breast histopathology images," Kaggle, San Francisco, CA, USA, 2020. Accessed: Dec. 2, 2024. [Online]. Available: https://www.kaggle.com/datasets/paultimothymooney/breast-histopathology-images
[45] M. Tan and Q. V. Le, "Efficientnetv2: Smaller models and faster training," in *Proc. 38th Int. Conf. Mach. Learn.*, Oct. 2021, pp. 10096–10106.

WISHAL ARSHAD received the B.S. and M.S. degrees in computer science from UCP Lahore. She is currently pursuing the Ph.D. degree with Superior University, Lahore. She is actively engaged in academia and is a Faculty Member with the Software Engineering Department, UCP Lahore. With a strong academic background and dedication to teaching and research, she contributes significantly to the academic community. Her research interests primarily revolve around machine learning and deep learning, where she explores innovative approaches to advance the field.



TEHREEM MASOOD received the bachelor's degree (Hons.) in software engineering from Punjab University, the M.S. (Research) degree (Hons.) in software engineering from MAJU Islamabad, and the Ph.D. degree in computing from the Decision and Information Systems for Production Systems (DISP) Laboratory, INSA Lyon, University de Lyon, France.

She is currently an Academician, a Software Engineer, a Data Scientist, a Researcher, and an Entrepreneur with leadership qualities. With 14 years of experience spanning industry, academia, and research at both national and international levels, she is a Senior Member of Intelligent Data Visual Computing Research (IDVCR). She is also an HEC-approved Ph.D. Supervisor. Previously, she held the position of a Senior Software Engineer in a research and development setup with Pakistan Air Force. She was a recipient of the Erasmus Mundus Scholarship under the Clink Project, France. Her research interests include data science, machine learning, deep learning, software testing, semantic web, and service-oriented architecture. She has implemented performance-based semantic, analytical, and predictive models for service sustainability and decision support in SOA following a machine learning approach. These models were validated with the implementation of an industrial business process use case and public dataset repositories of shared services. Additionally, she developed the RWest Tool for specification-based regression testing of web services.

H. M. SHAHZAD received the master's degree from the COMSATS Institute of Information Technology, Lahore, Pakistan, and the Ph.D. degree in computer science from Superior University.

He is currently an Assistant Professor with the Software Engineering Department, Superior University, Lahore. He has more than 14 years of teaching experience at the university level. He is also actively involved in supervising undergraduate and graduate research projects in AI and deep learning. His current research interests include image processing, machine learning, deep learning, and computer vision. His research interests include data science, image processing, machine learning, and deep learning, with various publications in national and international journals.

HASSAN A. AHMED received the D.M. and D.B.A. degrees from Case Western Reserve University, Cleveland, OH, USA, and the M.B.A. degree from Kent State University, Kent, OH.

He is currently a College Lecturer with the Information Systems Department, Cleveland State University, Cleveland. He has over 20 years of experience in lean manufacturing within the food industry and supply chain management. He also serves as the Faculty Advisor for Cleveland Chapter of the Association of Information Systems at Cleveland State University. His current research interests include business performance, business intelligence, information systems, operations management, supply chain management, systems thinking, system analysis, project management and dynamic modeling for complex systems.

SYED HAMZA AHMED received the Bachelor of Science degree in electronics from COMSATS University Islamabad, Islamabad, Pakistan, and the master's degree in information systems from Cleveland State University, Cleveland, OH, USA.

He brings nearly four years of experience in data engineering and development. His expertise spans ETL processes, business intelligence, and advanced data systems. With research interests in data engineering, information systems, ETLs, cybersecurity, business intelligence, signal and image processing, and neural networks, he is committed to leveraging data-driven solutions to address complex challenges.

HAFIZ MUHAMMAD TAYYAB KHUSHI received the B.S. degree in computer science from Bahauddin Zakariya University, Multan, in 2018, and the M.S. and Ph.D. degrees in computer science from Superior University, Lahore, in 2020 and 2025, respectively.

His research interests include data science, image processing, machine learning, and deep learning, with various publications in national and international journals. He is a reviewer of Springer Nature.